AGENDA

About Hypervisor.

About VMware Hypervisor.

- ESX Hypervisor Vs ESXi Hypervisor.
- ESXi Hypervisor Vs Hyper-V Hypervisor.

Installation of ESXi 8.0 - LAB

Customization of Hypervisor ISO - LAB

About DCUI Configuration - LAB

About Booting Process of ESXi Host

About VMware vSphere Client

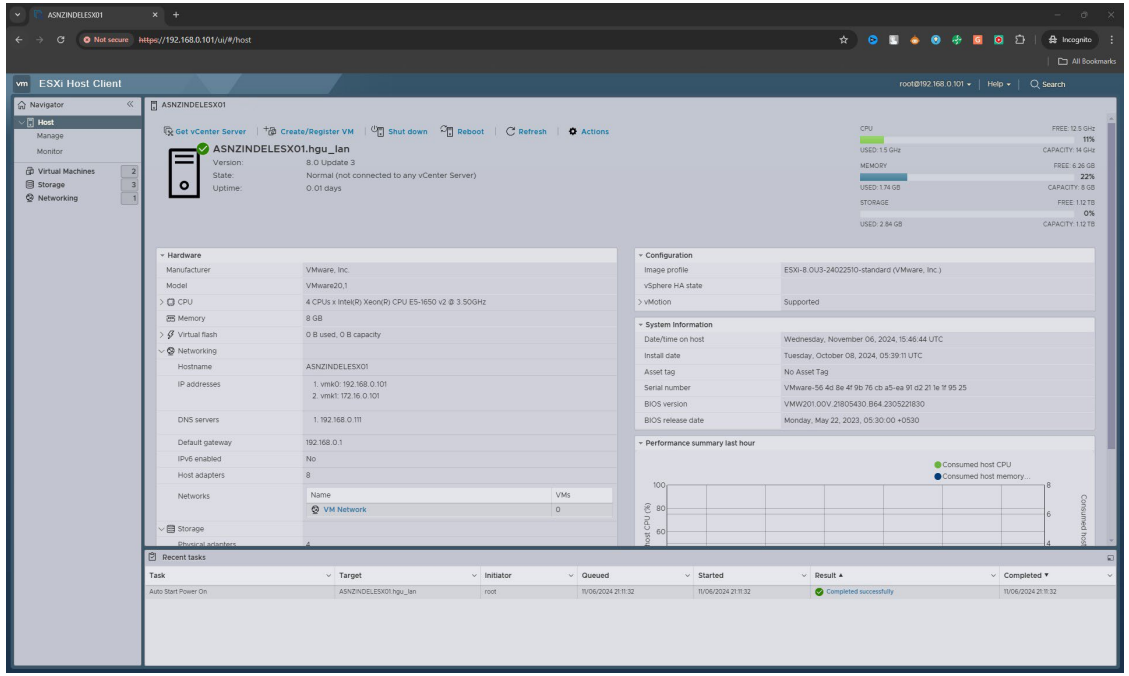
Installation of vSphere Client - LAB

Connecting ESXi Hypervisor - LAB

Licensing of ESXi Hypervisor – Theory | LAB

Services of ESXi Hypervisor – Theory | LAB

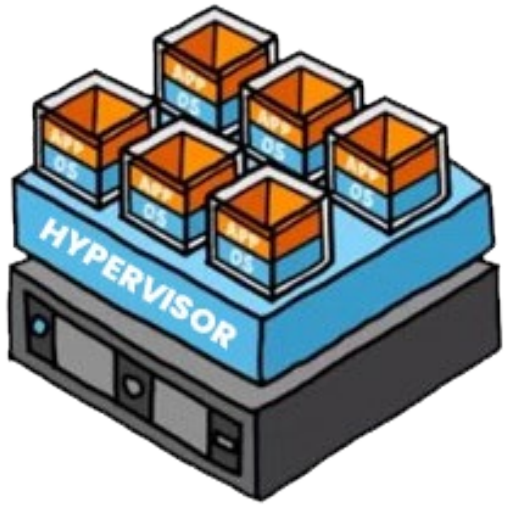
Upgradation of ESXi Hypervisor 7.0 to ESXi Hypervisor 8.0 - LAB



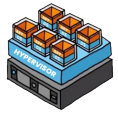


VMware ESXi
HYPERVISOR

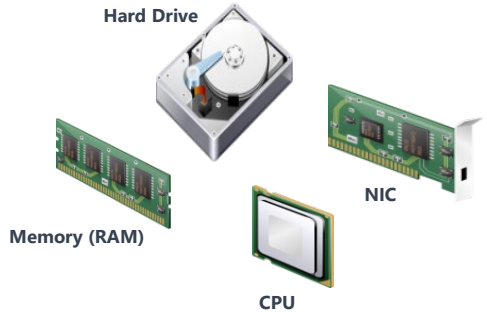
HYPERVISOR



HYPERVISOR

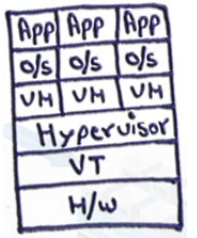


A hypervisor is a hardware virtualization technique that allows multiple guest operating system (Virtual Machine) to run on a single host system at the same time.
Virtual Machine shares the hardware of the host computer, such that each operating system appears to have its own processor, memory, storage and other hardware resources.
A hypervisor is also known as Virtual Machine Monitor (VMM) is a piece of computer software, firmware or hardware that creates and runs virtual machines.



Type 1 Hypervisor (Bare-Metal)

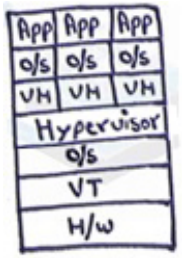
- Bare Metal hypervisors run directly on the host's hardware to control the hardware and to manage virtual machine. For this reason, they are sometimes called bare metal hypervisors. The first hypervisors, which IBM developed in the 1960s, were native hypervisors for testing software SIMMON and the CP/CMS operating system. Some Native / Bare-Metal hypervisors are Citrix, Xen Server, VMware ESX/ESXi and Microsoft Hyper-V 2008/2012.



Type 2 Hypervisor (Hosted | Para Hypervisor)

- These hypervisors run on a conventional operating system just as other computer programs do. Hosted hypervisors abstract virtual machine from the host operating system. VMware Workstation, VMware Player, Virtual Box and QEMU are examples of type-2 hypervisors.





Benefits:
- Better hardware utilization.
- Isolation of applications and environments.
- Easier backup, migration, and scaling.
- Running multiple OSes on the same physical server.

VMware ESXi
HYPERVISOR

VMware ESXi 8.0
VMVISIOR



ESXi HYPERVISOR 8.0

VMware vSphere 8.0 | 11 OCT 2022
ESXi 8.0 | 11 OCT 2022 | Build 20513097
vCenter Server 8.0 | 11 OCT 2022 | Build 20519528
Check for additions and updates to these release notes.

VMware ESXi 8.0 is the latest version of VMware's enterprise-grade, **Type 1 hypervisor** that enables virtualization of physical hardware. ESXi is designed to run directly on the physical hardware of a server without needing a host operating system, offering high performance, scalability, and reliability for virtualized environments.

Here are the key features and updates of **VMware ESXi 8.0:**

Improved Performance and Scalability

- Support for More CPUs and Memory: ESXi 8.0 enhances support for larger virtual machines (VMs) and hosts, allowing up to 128 vCPUs per VM and 6 TB of RAM.
- CPU and Memory Optimizations: ESXi 8.0 improves resource management, making it more efficient at utilizing underlying hardware. This leads to better performance and scalability in large deployments, particularly in cloud, AI, and big data workloads.

Enhanced Security Features

- Secure Boot & TPM 2.0: ESXi 8.0 extends support for Trusted Platform Module (TPM 2.0) and Secure Boot for improved security of both the host and virtual machines.
- vSphere Security Updates: Regular security patches and updates to ensure compliance with industry standards and best practices.
- VM Encryption: It includes further advancements in VM encryption, protecting data both in transit and at rest.



vSphere Lifecycle Management

- Simplified Update and Upgrade Processes: ESXi 8.0 improves on the vSphere Lifecycle Manager (vLCM), simplifying firmware and software updates across your virtual infrastructure.
- Offline Bundle Support: New capabilities around offline updates, making it easier to patch or update ESXi hosts without internet connectivity.

INTRODUCTION OF COURSE

ESXi HYPERVISOR 8.0

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Here are the key features and updates of **VMware ESXi 8.0:**

Better Support for Modern Hardware

- New Hardware Compatibility: ESXi 8.0 provides expanded support for the latest server hardware, including new processors, GPUs, and storage devices, allowing you to run resource-intensive workloads like AI and machine learning applications more effectively.
- Hardware Accelerated Networking: ESXi 8.0 integrates better with modern networking hardware, providing hardware offloading capabilities to enhance network performance for virtualized workloads.

VMware vSphere Distributed Switch (vDS) Enhancements

- ESXi 8.0 improves network management and monitoring by providing advanced features for managing virtual switches and enhanced troubleshooting tools for network performance issues in virtualized environments.
- Support for Network Automation: Better integration with automation tools like Ansible or vRealize Automation, allowing network policies to be managed and deployed more efficiently.

vMotion and Live Migration Improvements

- Enhanced vMotion Capabilities: ESXi 8.0 delivers faster and more efficient vMotion (live migration of VMs between hosts), reducing the overhead involved and improving performance.
- Cross-Cluster vMotion: Easier movement of VMs between clusters in multi-tenant environments.



vSphere AI and GPU Support

- NVIDIA vGPU and GPU Passthrough: ESXi 8.0 has better support for GPU virtualization, allowing virtual machines to access GPU resources for workloads like machine learning, high-performance computing (HPC), and rendering.
- AI Workload Optimization: Better optimizations for AI-driven workloads and integration with platforms like NVIDIA AI Enterprise.
- New Storage and Data Management Features
- vSAN Enhancements: VMware vSAN, which integrates tightly with ESXi, has improvements in data management, redundancy, and storage efficiency.
- NVMe over Fabrics Support: Improved support for NVMe storage technologies, offering faster data access for virtualized environments.
- Storage Policies: Enhanced storage policies for VMs, with more granular controls and automated storage tiering.

ESXi HYPERVISOR 8.0

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Here are the key features and updates of **VMware ESXi 8.0:**

Support for Containers and Kubernetes

- vSphere with Tanzu: ESXi 8.0 continues to integrate closely with VMware Tanzu, making it easier to deploy and manage containerized workloads alongside traditional virtual machines.
- VMware NSX-T Integration: For software-defined networking (SDN) across containers, ESXi 8.0 works more seamlessly with NSX-T, enabling container networking and service mesh capabilities.

Cloud Integration and Multi-cloud Support

- ESXi 8.0 improves integration with VMware Cloud Foundation and various public cloud services, allowing for hybrid cloud environments that can span both on-premises and cloud-based resources.
- VMware vSphere+: Integration with vSphere+ allows customers to manage ESXi hosts from the cloud, offering simplified management for hybrid infrastructures.

Support for Nested Virtualization

- ESXi 8.0 continues to support nested virtualization, which allows running ESXi as a VM on top of another ESXi host. This is useful for testing and lab environments, particularly when you want to simulate large-scale environments without needing the corresponding physical hardware.



INTRODUCTION OF COURSE

ESXi HYPERVISOR 8.0

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Benefits of ESXi: -

- Decrease hardware, power and cooling costs by running multiple operating systems on the same physical server.
- Lower management overhead costs by reducing the hardware footprint in the datacenter.
- Guarantee high levels of performance for the most resource-intensive applications.
- Consolidate hardware resources with the peace of mind that comes with the industry's most widely deployed, production-proven and secure server virtualization platform.



Requirements for VMware ESXi 8.0:

- CPU: 64-bit x86 processors with hardware virtualization support (Intel VT-x or AMD-V).
- Memory: Minimum of 4 GB RAM (8 GB or more recommended).
- Storage: 5 GB or more of local storage for installation; additional storage for virtual machines.
- Network: At least one Gigabit Ethernet adapter (multiple adapters recommended for high availability and network redundancy).

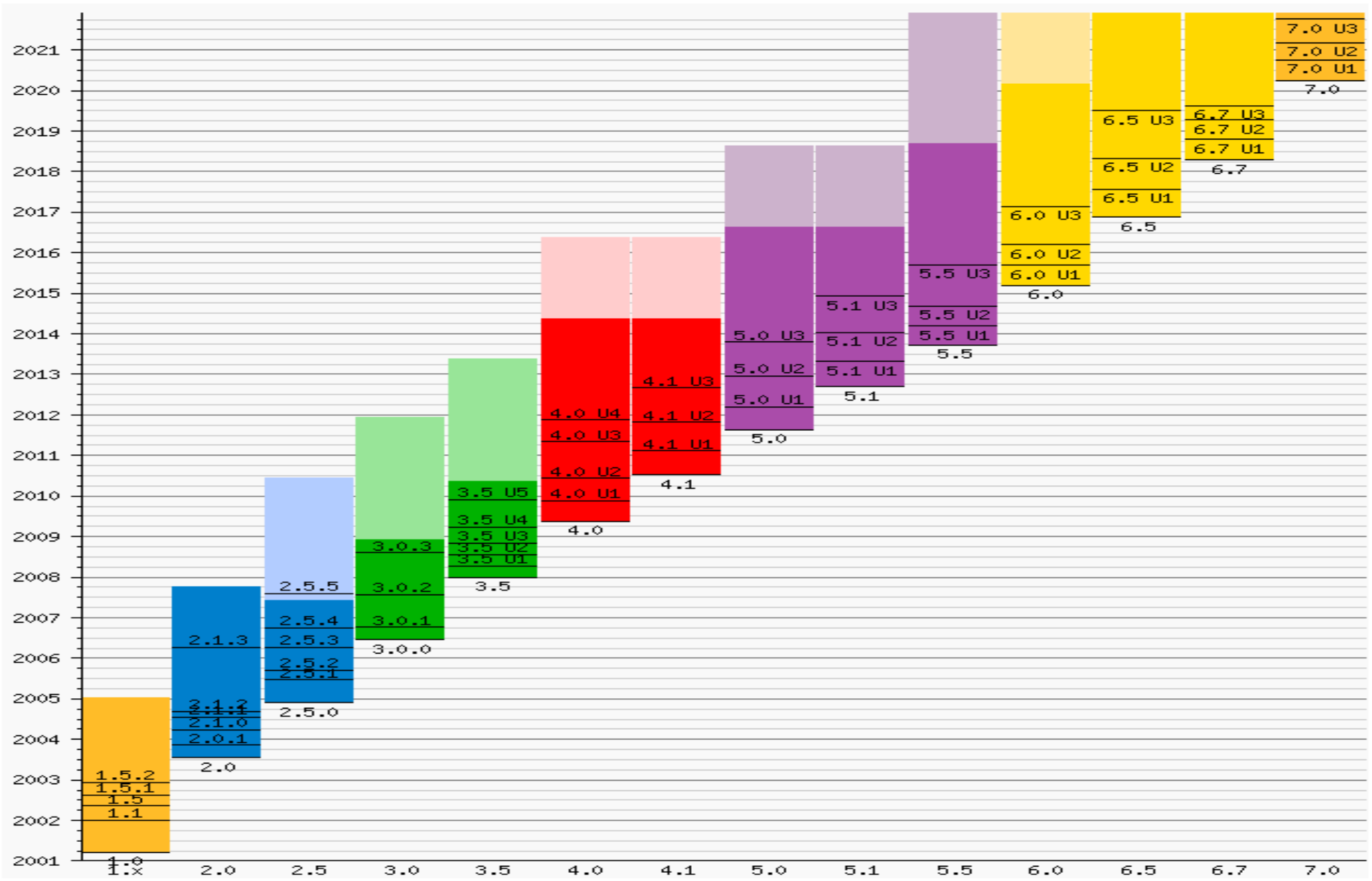








HYPERVISOR HISTORY



ESXi HYPERVISOR 8.0



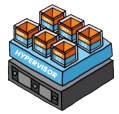
Edition's of VMware ESXi 8.0:

- Standard
- Enterprise
- Enterprise Plus
- Tanzu



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HYPERVISOR VARIATION



	vSphere 4.0	vSphere 4.1	vSphere 5.0	vSphere 5.1	vSphere 5.5	vSphere 6.0	vSphere 6.5	vSphere 6.7	vSphere 7.0
Released	05/2009	07/2010	08/2011	09/2012	09/2013	03/2015	11/2016	04/2018	04/2020
Hypervisor	ESX & ESXi	ESX & ESXi	ESXi	ESXi	ESXi	ESXi	ESXi	ESXi	ESXi
CPUs per host	64	160	160	160	320	480	576	768	768
RAM per host	1 TB	1 TB	2 TB	2 TB	4 TB	12 TB	12 TB	16 TB	16 TB
vCPU per VM	8	8	32	64	64	128	128	128	128
vMEM per VM	255 GB	255 GB	1 TB	1 TB	1 TB	4 TB	6 TB	6 TB	6 TB
VMDK Size	2TB	2TB	2TB	2TB	62 TB	62 TB	62 TB	62 TB	62 TB
Cluster Size	32	32	32	32	32	64	64	64	64
FT Limit	1 vCPU 64 GB	1 vCPU 64 GB	1 vCPU 64 GB	1 vCPU 64 GB	1 vCPU 64 GB	4 vCPU 64 GB	4 vCPU 64 GB	8 vCPU 128 GB	8 vCPU 128 GB
VM Hardware Version	7	7	8	9	10	11	13	14+15	17
VMFS Version	3.33	3.46	5.54	5.58	5.60	5.61	5.81 (VMFS5) 6.81 (VMFS6)	5.81 (VMFS5) 6.82 (VMFS6)	5.81 (VMFS5) 6.82 (VMFS6)
New Features	- Host Profiles - Storage vMotion - Update Manager	- Boot from SAN - Network I/O Control	- Storage DRS - vSphere Web Client - ESXi Firewall	- vSphere Data Protection - Enhanced vMotion	- Virtual SAN - vCenter Single Sign-On 2.0	- Virtual Volumes - vMotion across vSwitches	- vCenter Migration - vCenter Server High Availability	- Single Reboot Upgrade - Instant Clone	- Tanzu - Trust Authority - Lifecycle Manager - Improved DRS

ESXi HYPERVISOR 8.0

ESX (Elastic Sky X) is the VMware's enterprise server virtualization platform. In ESX, VMKernel is the virtualization kernel which is managed by a console operating system which is also called as Service Console. Which is Linux based and its main purpose is it to provide a management interface for the host and lot of management agents and other third-party software agents are installed on the Service Console to provide the functionalists like hardware monitor and management of ESX hypervisor.

ESXi (Elastic Sky X Integrated) is also the VMware's enterprise server virtualization platform. In ESXi, Service Console is removed. All the VMware related agents and third party agents such as management and monitoring agents can also run directly on the VMKernel. ESXi is ultra-thin architecture which is highly reliable and its small code base allows it to be more secure with less code patch. ESXi uses Direct Console User Interface (DCUI) instead of a Service Console to perform management of ESXi server. ESXi installation will happen very quickly as compared to ESX installation.



Capability	ESX	ESXi
Service Console	Present	Removed
Troubleshooting performed via	Service Console	ESXi Shell
Active Director Authentication	Enabled	Enabled
Secure Syslog	Not Supported	Supported
Management Network	Service Console Interface	VMKernel Interface
Jumbo Frames	Supported	Supported
Hardware Montioring	3 rd Party agents installed in Service console	Via CIM Providers
Boot from SAN	Supported in ESX	Supported in ESXi
Software patches and updates	Needed as smilar to linux operation system	Few pacthes because of small footprint and more secure
vSphere web Access	Only experimental	Full managenet capability via vSPhere web client
Locked Down Mode	Not present	Present . Lockdown mode prevents remote users to login to the host
Scripted Installtion	Supported	Supported
vMA Support	Yes	Yes
Major Administration command-line Command	esxcfg-	esxcli
Rapid deployment via Auto Deploy	Not supported	Supported
Custom Image creation	Not supported	Supported
VMkernel Network Used for	vMotion,Fault Tolarance,Stoarge Connectivity	Management Network , vMotion, Fault Tolarance, Stoarge Connectivity, ISCSI port binding